ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 13	3n + 1 = 13.
2	**(a)** $CH_2{=}CH_2$	Dehydration.
3	**(d)** forms ionic bonds with chlorine	Carbon forms covalent bonds.
4	**(d)** 10	3 x 3 + 1 = 10.
5	**(d)** CH_3COOH	Oxidation.
6	**(d)** high energy is needed to remove 4 electrons	Too much energy.
7	**(c)** 6	4 C-H pairs plus 2 pairs in C=C.
8	**(b)** C_2H_4	Unsaturated compounds.
9	**(d)** 6 protons cannot hold 10 electrons easily	Nucleus can't hold extra electrons.
10	**(b)** 5	2 C-H plus 3 in triple bond.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Needs 4 electrons to complete octet (1); Loss/gain requires too much energy (1).	1+1
12	Add bromine water (1); Unsaturated decolourises it (1).	1+1
13	$HCOOH + NaOH \longrightarrow HCOONa + H_2O$ (2).	2
14	Covalent: low mp, poor conductors (1); Ionic: high mp, conduct when molten/dissolved (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Formula 3n + 1 (1); Hexane: 19 (1); Decane: 31 (1).	1+1+1
16	Soap: fatty acid salts; detergent: sulphonate salts (1); Soap forms scum in hard water (1); Detergents don't (1).	1+1+1
17	Sharing of electron pairs (1); Cl_2: single bond (1); O_2: double bond (1).	1+1+1
	OR: Diamond has 3D arrangement (1); Graphite has layers (1); Free electrons in graphite make it conductor (1).	1+1+1
18	Hydrophobic tail avoids water (1); Heads face water (1); Cluster forms a sphere (micelle) (1).	1+1+1
	OR: Ethanoic acid is weak, HCl strong (1); Less H^+ (1); Slower reaction with metals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tetravalency: four bonds (1); Catenation: long chains, branches, rings (1); Multiple bonds possible (1); Isomerism (1); Strong C-C bonds (1).	1+1+1+1+1
	OR: $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (sunlight) (1); $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (Ni) (1); $CH_2{=}CH_2 + Br_2 \longrightarrow CH_2Br{-}CH_2Br$ (1); Bromine water decolourised (1); Addition for unsaturated (1).	1+1+1+1+1
20	C_2H_6, C_2H_4, C_2H_2 (1); Single, double, triple bonds (1); Ethane clean flame (1); Ethene and ethyne sooty (1); Ethene and ethyne show addition (1).	1+1+1+1+1
	OR: Methane: 4 C-H (1); Ethane: 6 C-H, 1 C-C (1); Propane: 8 C-H, 2 C-C (1); Butane: 10 C-H, 3 C-C (1); Pentane: 12, 4; hexane: 14, 5 (1).	1+1+1+1+1

Mistakes to avoid:

- Counting covalent bonds wrongly: an alkane C_nH_{2n+2} has $3n + 1$ bonds (include all C-H and C-C bonds).
- Confusing ethanol and ethanoic acid: ethanol (C_2H_5OH) is an alcohol, ethanoic acid (CH_3COOH) releases CO_2 with carbonates; ethanol does not.
- Naming the chain from the wrong carbon count, e.g. calling CH_3CH_2CHO 'propanol' instead of propanal; the suffix comes from the functional group (-ol, -al, -one, -oic acid).
- Writing an addition reaction for a saturated hydrocarbon; saturated compounds undergo substitution, unsaturated ones undergo addition.
- Saying that detergents form micelles but soaps do not, or that soap works well in hard water; soaps give scum in hard water.
- Stating that diamond and graphite conduct electricity; only graphite does (it has free electrons).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** addition reaction	Double bond opens.
2	**(a)** no free electrons	All valence electrons used in bonds.
3	**(a)** 72 u	C_5H_{12} = 72.
4	**(b)** substitution reaction	H replaced by Cl.
5	**(a)** water	Soap micelles form in water.
6	**(d)** C_8H_{18}	$CnH_2n_2^+$.
7	**(a)** methyl ethanoate	Alcohol part first.
8	**(c)** inward towards oil	Tails attach to the grease.
9	**(a)** C_5H_{10}	CnH_2n.
10	**(a)** sodium ethanoate and ethanol	Alkaline hydrolysis.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Same functional group, same general formula (1); Successive members differ by CH_2 (1).	1+1
12	Forms insoluble scum with Ca/Mg ions (1); Reduces lather (1).	1+1
13	CH_3-CH_3 with C-C pair (1); 7 shared pairs (1).	1+1
14	Ethanol has low carbon percentage (1); Naphthalene is rich in carbon: incomplete combustion (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	n-pentane (1); 2-methylbutane (1); 2,2-dimethylpropane (1).	1+1+1
16	$CH_3COOC_2H_5 + NaOH \longrightarrow CH_3COONa + C_2H_5OH$ (2); Products: soap-like salt and ethanol (1).	2+1
17	Ethanol lowers metabolism (1); Methanol in denatured alcohol is poisonous (1); Added to prevent misuse (1).	1+1+1
	OR: A is ethanol (1); $2\,C_2H_5OH + 2\,Na \longrightarrow 2\,C_2H_5ONa + H_2$ (2).	1+2
18	Converts unsaturated oils into saturated fats (1); Nickel catalyst (1); Makes vanaspati ghee (1).	1+1+1
	OR: $C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O$ (2); $C_2H_6 + Cl_2 \longrightarrow C_2H_5Cl + HCl$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Conc. H_2SO_4, 443 K: ethene (1); Alkaline $KMnO_4$: ethanoic acid (1); Conc. H_2SO_4 catalyst: ester (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1); Conditions of each stated (1).	1+1+1+1+1
	OR: Propanol (1); Propanal (1); Propanoic acid (1); 1-chloropropane (1); Butanone (1).	1+1+1+1+1
20	Soap: fatty acid salts; detergent: sulphonate (1); Micelle formation (1); Hard water scum with soap (1); Detergents work in hard water (1); Detergents may be non-biodegradable (1).	1+1+1+1+1
	OR: Both react with Na (1); Only acid reacts with carbonate (1); Acid turns blue litmus red (1); Ethanol is neutral (1); Acid pH below 7 (1).	1+1+1+1+1

Mistakes to avoid:

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- Confusing ethanol and ethanoic acid: ethanol (C_2H_5OH) is an alcohol, ethanoic acid (CH_3COOH) releases CO_2 with carbonates; ethanol does not.
- Naming the chain from the wrong carbon count, e.g. calling CH_3CH_2CHO 'propanol' instead of propanal; the suffix comes from the functional group (-ol, -al, -one, -oic acid).
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** it can form hydrogen bonds	-OH group.
2	**(b)** C_4H_6	CnH_2n-2.
3	**(d)** ethanoic acid	Acids react with both.
4	**(c)** water	Strong hydrogen bonding.
5	**(d)** butanoic acid	Four carbon acid.
6	**(a)** ethanol	Alcohol.
7	**(c)** C_2H_5COOH	Adds CH_2.
8	**(a)** propanone	Ketone.
9	**(c)** hydrogenation	Ni catalyst.
10	**(b)** 2	Butane and isobutane.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Propanoic acid (1); C_2H_5COOH (1).	1+1
12	It is saturated with single bonds only (1); No double bond to open (1).	1+1
13	Acts as a catalyst (1); Also removes water formed (1).	1+1
14	$C_2H_5OH + 3\,O_2 \longrightarrow 2\,CO_2 + 3\,H_2O$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	X: ethanoic acid (1); Gas: CO_2 (1); $CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2$ (1).	1+1+1
16	Alcohol molecules form hydrogen bonds (1); More energy needed to separate them (1); Hydrocarbons only have weak forces (1).	1+1+1
17	No hydrogen and fully halogenated (1); Already oxidised, no combustion (1); Used in fire extinguishers (1).	1+1+1
	OR: Incomplete combustion gives CO (1); CO is poisonous (1); Binds to haemoglobin (1).	1+1+1
18	-CHO aldehyde: C_2H_5CHO (1); -OH alcohol: C_3H_7OH (1); Add CH_2 each time (1).	1+1+1
	OR: Ethanoic acid turns blue litmus red (1); Ethanol is neutral (1); Also by bicarbonate test (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A = ethanol (1); B = ethanoic acid (1); C = ethyl ethanoate (1); $CH_3CH_2OH \longrightarrow CH_3COOH$ (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1).	1+1+1+1+1
	OR: Fuel: LPG (1); Soap: sodium stearate (1); Solvent: ethanol (1); Medicine: ethanol tinctures (1); Plastics: polythene (1).	1+1+1+1+1
20	Diamond: rigid 3D, hard, no free electrons (1); Graphite: layers, conducts (1); Soft because layers slide (1); Fullerene: C_{60} sphere (1); Uses relate to property (1).	1+1+1+1+1
	OR: Low mp due to weak intermolecular forces (1); Poor conductors: no ions (1); Insoluble in water usually (1); Soluble in organic solvents (1); Examples: methane, ethanol (1).	1+1+1+1+1

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